CO-1 ⇒ AT-1

Date:- 16/10/26

Course name : Operating System.
Course code : CSA0417
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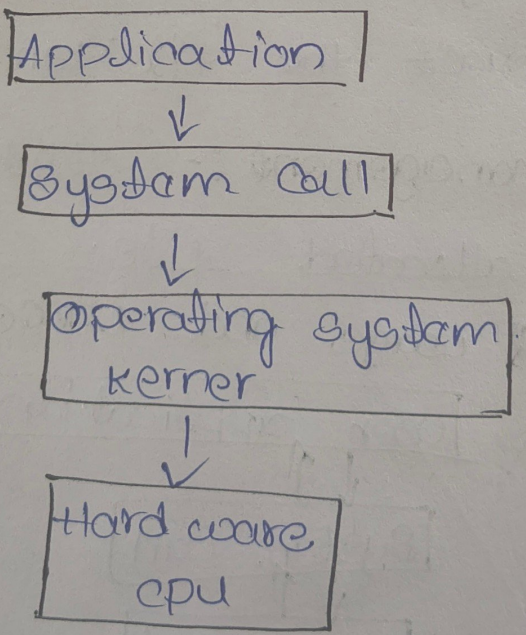
1) a) modern operating system.
b) different type of operating system.

i) Multiprogram operating system :-
keep multiple programs in memory to improve the cpu utilization.

ii) Multi task operating system :-
flow multiple or many processes.

iii) Time sharing operating system :
use or allocate cpu Time to different processes.

b) Operating System structure :-



c) System calls :-

* fork() :

* exec(): use to execute program.
* wait(): parent ware until the child.
* exit(): Terminate the program.

* fork() → exec() → wait() → exit()

2) a) System call usage for file and Device management:

i) file management :- open(). read(). write(), close(). unlink()

ii) Device management :- roct(), read()

communication.

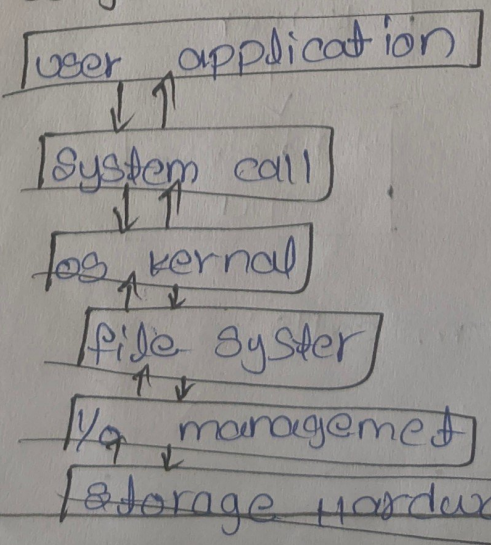
iii) process management : fork(), exit()

create and correct the file) terminate the program.

iv) Information management :- start().

Information allocated file.

b) System call usage and Device management diagram.



Realizatio

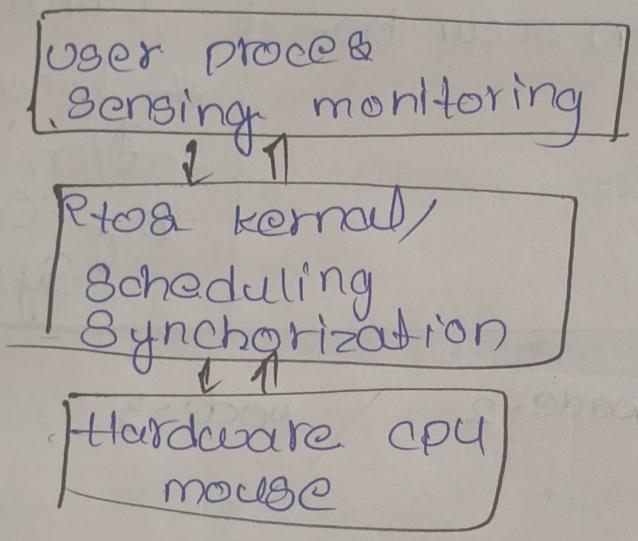
Reduction of operating system (RTOS) and for containing the industry machanism.

-> provide immediate respones

-> Report to interup quickly

-> Scheduling high priority task.

b)



c) System call for process Synchorization.

- Sem wait() - wait for Synchorization before access to the shared.
- Sem post() - signal semphore and allow another task to proceed.
- pthread . mutex . lock() lock shared resources.
- Pthread . mutex . unlock(). Release the locked resource.

4) A Distributed operating System ma... gas multiple interconnect Computer and... Coordinat resource.

Distributed O.S	Central O.S
* Use multiple interconnect Computer * Resources are distributed * Communication occur thro-gh network.	* Mainly manages by Central Computer * Resources are Centralized * local Communi-cation manage by Central System.

b)

node 1	node-2	node-3
cpu+	–	–
mem+	–	–
Resources	Through	network

c) socket() : create Communicate end point.
bind() : Assign address no. to a socket
listen() : waits are incoming Connection.
Accept() : Accept Connect.
close() : close Communication.

equation : Socket() → bind(), listen() → accept()
→ send() / recv() .. close().

major function of an operating system:

i) process management :-
creates, schedules and terminating the processes.

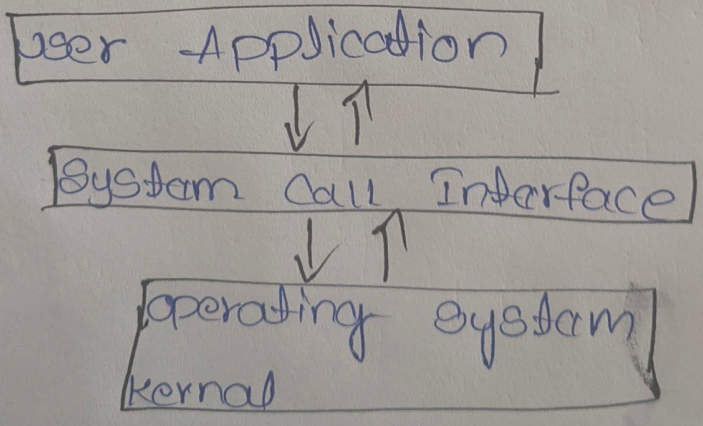
ii) memory management :-
Allocates and deallocates memory and manages virtual (images) memory.

iii) Device management :-
controls I/O devices through device drivers.

iv) file management :-
Organizes, stores, retrives, and protects files.

v) Resource allocation :-
Distribute cpu time, memory and devices efficiently.

b) block diagram of operating system.



c) System call ex for memory allocation.

Memory allocation :-

- mmap() - maps memory into a process's address space.
- munmap() - Releases a memory mapping.
- brk()/sbrk() - Adjusts the process heap boundary on system that support them.

Device management :-

- open() - opens a device file.
- read() - reads data from a device.
- write() - Sends data to a device
- ioctl() - performs device-specific Control Operations.
- close() - closes the device file.